

YEAR 1

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 2

ORGANIZATION CUNY Graduate School University Center				FOR NSF USE ONLY		
				PROPOSAL NO.	DURATION (months)	
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Gary Welz				AWARD NO.	Proposed	Granted
A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/Key Associates (List each separately with title, A.7. show number in brackets)		NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)	
		CAL	ACAD	SUMR		
1. Gary Welz - Principal Inv		6.0			60,000	
2.						
3.						
4.						
5.						
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)		0.0			0	
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)		6.0			60,000	
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)						
1. (0) POST DOCTORAL SCHOLARS		0.0			0	
2. (1) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)		4.0			20,000	
3. (2) GRADUATE STUDENTS					6,000	
4. (0) UNDERGRADUATE STUDENTS					0	
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)					0	
6. (0) OTHER					0	
TOTAL SALARIES AND WAGES (A + B)					86,000	
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)					0	
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)					86,000	
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)						
High-performance dev laptop, dev tools, software				\$ 5000		
Monitoring + analytics tools, and cloud service credits/reserv				\$ 3000		
TOTAL EQUIPMENT					8,000	
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)					4,000	
2. INTERNATIONAL					0	
F. PARTICIPANT SUPPORT COSTS						
1. STIPENDS \$ _____		0				
2. TRAVEL _____		0				
3. SUBSISTENCE _____		0				
4. OTHER _____		0				
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS					0	
G. OTHER DIRECT COSTS						
1. MATERIALS AND SUPPLIES					5,000	
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION					3,000	
3. CONSULTANT SERVICES					20,000	
4. COMPUTER SERVICES					5,000	
5. SUBAWARDS					0	
6. OTHER					2,000	
TOTAL OTHER DIRECT COSTS					35,000	
H. TOTAL DIRECT COSTS (A THROUGH G)					133,000	
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) Facilities and Administrative (F&A) Costs (Rate: 20.0, Base:133000)						
TOTAL INDIRECT COSTS (F&A)					26,600	
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)					159,600	
K. FEE					0	
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)					159,600	
M. COST SHARING PROPOSED LEVEL \$ 0		AGREED LEVEL IF DIFFERENT \$				
PI/PD NAME Gary Welz		FOR NSF USE ONLY				
ORG. REP. NAME*		INDIRECT COST RATE VERIFICATION				
		Date Checked	Date Of Rate Sheet	Initials - ORG		

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 3

ORGANIZATION CUNY Graduate School University Center				FOR NSF USE ONLY			
				PROPOSAL NO.	DURATION (months)		
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Gary Welz				AWARD NO.			
A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/Key Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Gary Welz - Principal Inv				6.0			60,000
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0			0
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)				6.0			60,000
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.0			0
2. (1) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				4.0			20,000
3. (2) GRADUATE STUDENTS							6,000
4. (0) UNDERGRADUATE STUDENTS							0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							86,000
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							0
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							86,000
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
High-performance dev laptop, dev tools, software						\$ 5000	
Monitoring + analytics tools, and cloud service credits/reserv						\$ 3000	
TOTAL EQUIPMENT							8,000
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							4,000
2. INTERNATIONAL							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ _____							0
2. TRAVEL _____							0
3. SUBSISTENCE _____							0
4. OTHER _____							0
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							5,000
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION							3,000
3. CONSULTANT SERVICES							20,000
4. COMPUTER SERVICES							5,000
5. SUBAWARDS							0
6. OTHER							2,000
TOTAL OTHER DIRECT COSTS							35,000
H. TOTAL DIRECT COSTS (A THROUGH G)							133,000
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)							
Facilities and Administrative (F&A) Costs (Rate: 20.0, Base:133000)							
TOTAL INDIRECT COSTS (F&A)							26,600
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							159,600
K. FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							159,600
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PD NAME Gary Welz				FOR NSF USE ONLY			
				INDIRECT COST RATE VERIFICATION			
ORG. REP. NAME*				Date Checked	Date Of Rate Sheet	Initials - ORG	

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION CUNY Graduate School University Center				FOR NSF USE ONLY					
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Gary Welz				PROPOSAL NO.		DURATION (months)			
				AWARD NO.		Proposed	Granted		
A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/Key Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer		Funds granted by NSF (if different)	
				CAL	ACAD	SUMR			
1. Gary Welz - Principal Inv				18.0			180,000		
2.									
3.									
4.									
5.									
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)									
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)				18.0			180,000		
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)									
1. (0) POST DOCTORAL SCHOLARS				0.0			0		
2. (3) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				12.0			60,000		
3. (6) GRADUATE STUDENTS							18,000		
4. (0) UNDERGRADUATE STUDENTS							0		
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0		
6. (0) OTHER							0		
TOTAL SALARIES AND WAGES (A + B)							258,000		
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							0		
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							258,000		
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)									
TOTAL EQUIPMENT							24,000		
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							12,000		
2. INTERNATIONAL							0		
F. PARTICIPANT SUPPORT COSTS									
1. STIPENDS \$ <u>0</u>									
2. TRAVEL <u>0</u>									
3. SUBSISTENCE <u>0</u>									
4. OTHER <u>0</u>									
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0		
G. OTHER DIRECT COSTS									
1. MATERIALS AND SUPPLIES							15,000		
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION							9,000		
3. CONSULTANT SERVICES							60,000		
4. COMPUTER SERVICES							15,000		
5. SUBAWARDS							0		
6. OTHER							6,000		
TOTAL OTHER DIRECT COSTS							105,000		
H. TOTAL DIRECT COSTS (A THROUGH G)							399,000		
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)									
TOTAL INDIRECT COSTS (F&A)							79,800		
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							478,800		
K. FEE							0		
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							478,800		
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$					
PI/PD NAME Gary Welz				FOR NSF USE ONLY					
ORG. REP. NAME*				INDIRECT COST RATE VERIFICATION					
				Date Checked	Date Of Rate Sheet		Initials - ORG		

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

Budget Justification

NSF CISE CORE (IIS) - CopernicusAI Knowledge Engine

Principal Investigator: Gary Welz **Project Period:** 36 months (3 years)

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- GCP services (increased usage): \$20,000
 - Third-party AI services: \$6,000
 - Vercel hosting: \$1,000
 - Network/data transfer: \$1,000
 - Year 3: \$25,000
 - GCP services: \$18,000
 - Third-party AI services: \$5,000
 - Vercel hosting: \$1,000
 - Network/data transfer: \$1,000
 - **Cloud Infrastructure Total: \$78,000**

Consultants/Subcontracts: - Year 1: \$5,000 (Metadata schema design consultation) - Year 2: \$7,000 (Quality assurance consultant, UI/UX consultant) - Year 3: \$5,000 (Evaluation consultant) - **Consultants Total: \$17,000**

Materials & Supplies: - Year 1: \$2,000 (Office supplies, testing materials) - Year 2: \$2,000 (Evaluation materials, user study materials) - Year 3: \$1,000 (Documentation materials) - **Materials Total: \$5,000**

Publication & Documentation: - Year 1: \$1,000 (Technical documentation) - Year 2: \$1,000 (User guides, evaluation reports) - Year 3: \$2,000 (Final documentation, publication costs) - **Publication Total: \$4,000**

Other: - Year 1: \$2,000 (Professional services, contingency) - Year 2: \$2,000 (Professional services, contingency) - Year 3: \$2,000 (Professional services, contingency) - **Other Total: \$6,000**

Justification: Cloud infrastructure costs reflect production-scale development and increased usage during user studies. Multi-model LLM architecture requires access to multiple providers. Consultants provide specialized expertise in metadata design, quality assurance, and evaluation. Materials and publication costs support documentation and dissemination.

H. Indirect Costs (\$78,750 total)

Indirect Cost Rate: 20% of modified total direct costs (estimated)

- Year 1: \$25,500 (20% of \$127,500)
- Year 2: \$27,750 (20% of \$138,750)
- Year 3: \$25,500 (20% of \$127,500)
- **Total: \$78,750**

Justification: Standard indirect cost rate for independent researcher arrangement. Covers general business operations, administrative overhead, and facilities costs.

Budget Justification Summary

This budget supports a 3-year research project to develop and evaluate an AI-enabled assisted research platform. The budget allocates:

- **40% to Senior Personnel:** PI effort (6 months/year) for research leadership and technical development
- **27% to Other Personnel:** Research assistant/developer support plus student user testers for alpha/beta testing and collaborative validation
- **24% to Other Direct Costs:** Primarily cloud infrastructure (18% of total) for production-scale development and multi-model LLM services
- **18% to Indirect Costs:** Standard rate for independent researcher arrangement

Key Cost Drivers: 1. **Personnel (67% combined):** Research-intensive work requires substantial PI and support personnel effort, plus student user testing for usability evaluation and collaborative validation 2. **Cloud Infrastructure (17% of total):** Production GCP services and multi-model LLM API costs are essential for the research 3. **Travel (4%):** Necessary for NSF meetings, conferences, and user study site visits

Student User Testing Component: The student user testing allocation (\$80,160 over 3 years) supports a critical research component: collaborative validation and real-world usability testing. Students will use the platform to create research briefings, test workflows, and participate in collective verification of AI outputs. This approach grounds the research in actual use cases while establishing mechanisms for collaborative intelligence that improve system reliability.

Year-by-Year Variations: - **Year 1:** Higher equipment costs for initial setup - **Year 2:** Increased personnel and cloud costs as user studies begin - **Year 3:** Focus on evaluation and documentation, slightly lower cloud costs

This budget is consistent with typical NSF CISE Core awards and supports the research objectives outlined in the proposal. The allocation reflects the technical nature of the work, the need for production-scale infrastructure, and the importance of evaluation and dissemination activities.

Cost Sharing

No cost sharing is proposed for this project.

Budget Prepared By: Gary Welz, Principal Investigator

Total Direct Costs Requested: \$441,000

Total Indirect Costs: \$88,200 (estimated at 20% of modified total direct costs)

Total Request: \$529,200

Note: Final indirect cost rate subject to negotiation with NSF.